

Principles of Silviculture

Land Sciences

May, 2026

Principles of Silviculture (A10283)

Short Title:	Principles of Silviculture	
Faculty	Science and Computing	
Department	Land Sciences	
Cluster	Forestry	
Author	NMCCARTHY	
Credits	5	Level: Intermediate

Description of Module / Aims

This module will provide an understanding of the silvicultural theory and practices involved in controlling forest establishment, composition and growth. This will include formulation of silvicultural systems for sites, and the development of an understanding of silvicultural practices, including natural and artificial regeneration, weed control regimes, nutrition and fertiliser, spacing, and intermediate stand treatments, such as thinning and pruning.

Programmes

FORS-0009 BSc in Forestry (WD_SFORS_D)

3 / 5 / M

Indicative Content

- Site preparation
- Choice of species
- Establishment and restocking
- Weed control
- Nutrition
- Plantation management practices: spacing, thinning, pruning
- Rotation: clearfelling

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use silvicultural terminology.
2. Examine principles of natural and artificial regeneration, spacing and intermediate stand treatments such as thinning and pruning.
3. Recommend methods of resolving problems associated with forest vegetation management.
4. Apply forest fertilisation principles to meet the nutrient requirements of trees and evaluate the potential environmental implications.
5. Apply silvicultural concepts such as species selection to simple case studies.
6. Collect ,collate and analyse results from silvicultural research trials.

Learning and Teaching Methods

- Lectures.
- Independent learning.
- Field trips.
- Case studies.
- Reports.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4
Continuous Assessment	50%	
<i>Practical</i>	30%	6
<i>Case Studies</i>	20%	5

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Forest Service, A. *_Code of best forest practice-Ireland_*. Ireland: Forest Service, 2000.
- Forest Service, A. *_Forestry Schemes Manual_*. Ireland: Forest Service, 2011.
- Savill, P., J. Evans, D. Auclair and J. Falck. *_Plantation Silviculture in Europe_*. UK: Oxford University Press, 1997.

Supplementary Material

- Pukkala , T. and K. Von Gadow. _Continuous Cover Forestry: Volume 23 (Managing Forest Ecosystems)_. 2nd ed. UK: Springer, 2013.
- Willoughby, I. _Creating new broadleaved woodlands_. UK: HMSO, 2004.

Requested Resources

- Lecture Room: Loose Seated